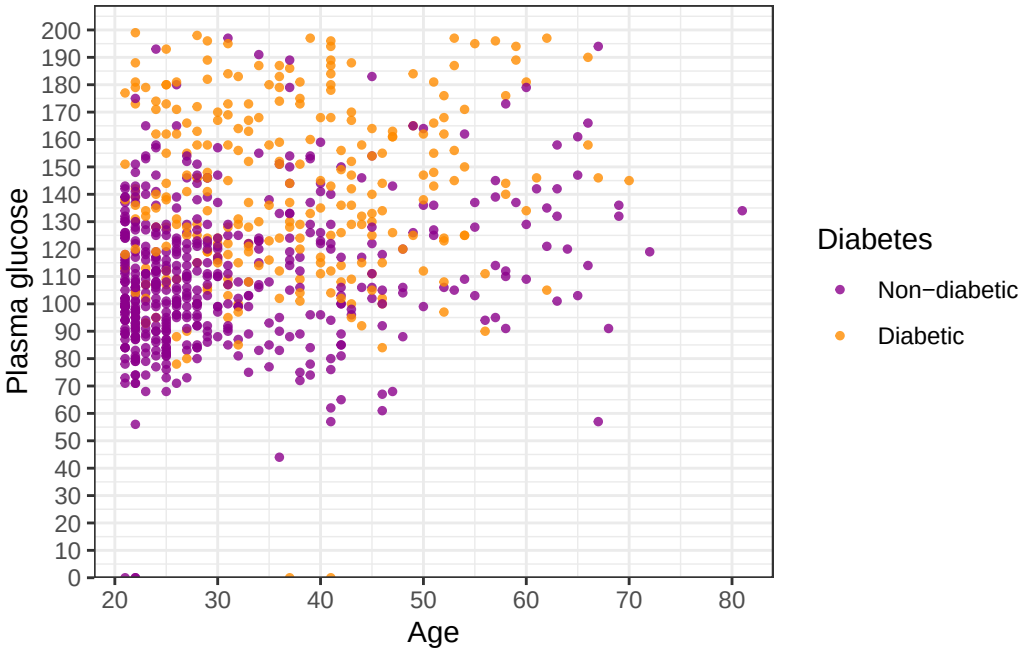
1 Assignment 3 - Logistic regression and Basis Function Expansion

1.1 Scatterplot diabetes vs age

```
plot_3_1
```



Plotting plasma glucose levels vs. age shows that the two response groups, `Non-diabetic` and `Diabetic`, show a substantial overlap. `Diabetes` would therefore not be easily predicted by `Age` and `Glucose` in a standard logistic regression model which would not be able to derive a straight line clearly separating the groups.

1.2 Logistic regression model

The vector form of the probailistic equation for a logistic regression model is

$$P(Y = 1 \mid \mathbf{x}) = \text{sigmoid}(\theta^T \mathbf{x}) \tag{1}$$

where the sigmoid function is

$$\text{sigmoid}(\theta^T \mathbf{x}) = \frac{1}{1 + e^{-\theta^T \mathbf{x}}} \tag{2}$$

For `Diabetes` as the binary target/response (y), and `Glucose` and `Age` as features/predictors (x1, x2):

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}} \tag{3}$$

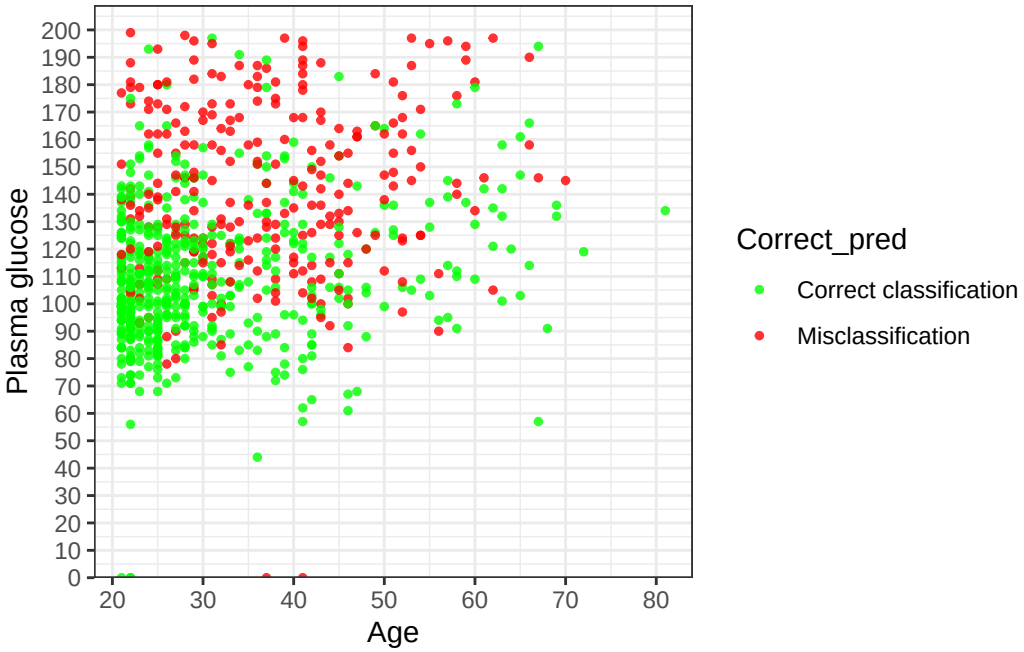
Training a logistic regression model on the (entire) provided data set, using glm() in R, results in the following probabilistic equation

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-5.8979 + 0.0356 x_1 + 0.0245 x_2)}} \tag{4}$$

The training misclassification error for this model is 0.27.

1.3 Scatterplot of predicted values

```
plot_3_2
```



1.4 Decision boundary equation

Setting the probabilistic equation Equation 4 = r (here = 0.5) and solving this gives the linear decision boundary equation as

$$-5.8979 + 0.0356x_1 + 0.0245x_2)$$